

BENTONVILLE, AR, USA

WMO: 723444

Lat: 36.350N

Lon: 94.217W

Elev: 1296

StdP: 14.02

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 03978

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.1	15.9	0.5	5.9	13.6	5.2	7.5	19.4	22.6	46.6	20.3	47.3	6.5	310	0.451

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.1	95.3	75.1	91.7	75.2	90.2	74.5	77.7	89.9	76.6	88.6	75.6	87.2	7.6	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.5	130.7	84.0	72.8	127.9	83.3	72.3	125.3	82.7	42.0	89.5	41.0	89.5	40.1	86.7	81.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.7	17.6	15.9	DB	3.7	97.3	4.7	2.8	0.3	99.3	-2.5	101.0	-5.1	102.6	-8.5	104.6	(4)
(5)			WB	2.5	79.2	4.5	1.3	-0.8	80.1	-3.4	80.8	-6.0	81.5	-9.2	82.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	58.3	36.2	40.3	49.0	57.7	66.3	74.8	78.7	78.1	70.6	59.4	48.6	39.2
	DBStd	17.15	11.15	10.96	10.28	8.62	7.46	5.40	4.72	5.28	7.11	8.84	9.30	10.03
	HDD50	1434	447	302	141	29	3	0	0	0	0	20	135	357
	HDD65	3953	894	691	500	245	77	4	0	1	30	217	495	799
	CDD50	4479	18	32	111	260	506	745	891	870	617	313	93	23
	CDD65	1522	0	0	5	26	116	299	426	405	197	45	3	0
	CDH74	14445	0	2	36	184	870	2749	4418	4097	1727	347	15	0
	CDH80	5860	0	0	2	16	189	1025	2003	1931	624	69	1	0
Wind	WSAvg	6.9	7.7	8.0	8.5	8.6	7.0	6.2	5.3	5.3	5.6	6.4	7.3	7.3
Precipitation	PrecAvg	45.4	2.5	2.2	3.6	4.4	5.9	4.7	3.8	2.8	4.3	3.3	4.3	3.1
	PrecMax	65.5	7.2	5.9	7.8	12.5	11.8	12.3	10.6	6.1	10.4	7.0	9.7	12.8
	PrecMin	32.4	0.2	0.3	0.8	1.5	1.5	1.1	0.9	0.0	0.5	1.0	0.6	0.3
	PrecStd	8.1	1.8	1.5	1.7	2.5	2.2	2.8	2.2	1.5	2.3	1.5	2.7	2.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	68.1	71.7	78.9	81.9	88.1	94.8	98.9	100.4	92.8	85.4	75.4	68.3
		MCWB	57.5	58.4	62.6	64.5	71.3	74.9	74.9	74.6	71.8	70.2	61.9	59.2
	2%	DB	63.4	66.0	73.2	79.2	84.2	90.8	94.6	97.0	90.0	81.7	71.7	64.0
		MCWB	54.7	55.1	59.3	63.5	70.2	74.8	75.7	74.5	72.4	67.4	59.4	56.1
	5%	DB	58.8	62.9	70.0	75.3	81.7	88.5	91.4	92.8	87.6	78.5	67.7	60.6
		MCWB	51.0	53.4	57.4	62.0	69.1	74.0	76.1	74.4	71.5	64.9	58.3	53.1
	10%	DB	54.3	58.6	65.7	72.6	79.3	86.0	90.1	90.2	83.5	73.3	63.9	54.9
		MCWB	47.0	50.0	55.7	60.5	67.7	73.3	75.5	74.0	70.1	63.3	55.5	48.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	60.5	61.5	64.7	69.0	74.8	77.9	79.1	78.9	75.8	72.8	65.7	61.4
		MCDB	65.5	68.0	75.8	78.1	83.6	89.7	90.4	90.9	87.6	81.7	74.0	66.8
	2%	WB	56.7	58.5	61.9	66.7	72.6	76.5	78.0	77.3	74.4	70.2	62.6	57.8
		MCDB	61.7	64.0	70.7	75.1	81.6	87.6	89.9	90.2	85.5	78.4	68.2	63.1
	5%	WB	52.4	54.6	59.4	64.8	70.8	75.3	77.0	76.2	73.0	67.9	59.8	54.0
		MCDB	57.4	60.7	66.9	72.6	79.6	85.7	89.0	88.9	83.2	74.5	65.7	58.6
	10%	WB	47.0	50.5	56.7	62.6	69.0	74.0	76.0	75.1	71.4	65.5	56.9	49.9
		MCDB	53.1	57.8	64.6	70.5	77.5	83.9	87.4	87.3	80.0	71.9	62.9	54.4
Mean Daily Temperature Range	5% DB	MDBR	18.9	19.5	20.5	21.2	19.1	18.4	19.1	20.4	20.6	21.3	20.2	18.1
		MCDBR	24.4	24.9	25.7	24.0	20.8	20.8	21.2	24.5	23.1	24.1	23.4	23.7
		MCWBR	16.8	17.2	15.6	13.5	10.8	9.1	8.0	8.5	9.4	12.6	14.9	16.7
	5% WB	MCDBR	20.4	21.3	21.5	19.3	18.5	18.8	19.5	20.3	19.2	19.0	20.4	19.8
		MCWBR	16.3	17.1	14.9	12.9	10.7	9.6	8.5	8.1	8.8	11.3	15.6	16.2
Clear-Sky Solar Irradiance	taub	0.302	0.305	0.339	0.376	0.420	0.421	0.442	0.443	0.386	0.343	0.311	0.291	
	taud	2.484	2.485	2.423	2.328	2.259	2.280	2.275	2.249	2.378	2.482	2.520	2.534	
	Ebn at Noon	285	298	295	287	275	273	267	264	275	279	279	282	
	Edh at Noon	26	29	34	40	43	43	42	43	36	29	25	23	
All-Sky Solar Radiation	RadAvg	778	948	1276	1610	1786	2042	2021	1834	1566	1163	836	662	
	RadStd	69	134	75	92	151	177	123	126	120	149	92	84	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (27 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page